

- 29** It was once thought that the mass of an atom is spread uniformly through the volume of the atom. When α -particles are directed at a piece of gold foil, the results led scientists to believe instead that nearly all the mass of the gold atom is concentrated at a point inside the atom.

Which effect is possible only if nearly all the mass of the gold atom is concentrated at a point?

- A** A few α -particles bounce back.
- B** Most α -particles are only slightly deflected.
- C** Some α -particles pass through without any deflection.
- D** Some α -particles are absorbed.

- 30** The isotope ^{222}Rn decays in a sequence of emissions to form the isotope ^{214}Po . At each